HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Led reliability improvement initiatives for critical Fill and Finish manufacturing equipment, focusing on filling lines and lyophilisers to reduce unplanned downtime.
- Developed and implemented **predictive and preventive maintenance strategies**, enhancing equipment uptime and Overall Equipment Effectiveness (OEE) across a fleet of 15+ assets.
- Conducted comprehensive **Root Cause Analysis (RCA)** and Failure Mode and Effects Analysis (FMEA) for recurring equipment failures, identifying key areas for improvement.
- Monitored and analyzed equipment performance metrics, including **Mean Time Between Failures (MTBF)** and Mean Time to Repair (MTTR), to inform maintenance planning.
- Supported the transition from reactive to **condition-based maintenance** approaches by identifying opportunities for sensor deployment and data analysis.
- Provided technical expertise and troubleshooting support for complex Fill and Finish equipment issues, resolving an average of **5 critical incidents per month**.
- Collected, analyzed, and reported on **reliability metrics and KPIs**, identifying equipment performance trends to support proactive maintenance interventions and data-based business cases.

SKILLS

Reliability Engineering – RCA, FMEA, RCM, MTBF, MTTR, OEE, Predictive Maintenance, Preventive Maintenance, Condition-Based Monitoring

Manufacturing Operations – Fill and Finish, Sterile Manufacturing, Aseptic Processing, GMP, cGMP, Regulatory Compliance (FDA, EMA)

Data Analysis & Tools – Microsoft Office Suite, Data Visualization Tools, Statistical Methods, Lean, Six Sigma

Maintenance Management – CMMS, Maintenance Strategies, Standard Operating Procedures (SOPs), Work Instructions

Technical Skills – Equipment Reliability, Asset Performance Optimization, Troubleshooting, Continuous Improvement Projects

PROJECTS

Fill Finish Equipment Reliability Enhancement

- Led a cross-functional team of 5 engineers and technicians to analyze failure data for critical lyophilisers and filling machines.
- Implemented a new **preventive maintenance schedule** based on criticality analysis, reducing unscheduled downtime by **18%** over a 6-month period.
- Developed and documented **Root Cause Analysis** procedures for common failure modes, improving incident response time by approximately **20%**.

OEE Improvement Initiative for Manufacturing Line

- Analyzed production data from a key filling line using **Microsoft Excel and Power BI** to identify key OEE loss factors.
- Developed and presented a business case for equipment upgrades based on data-driven insights, focusing on areas with the highest impact on **asset performance**.
- Collaborated with the maintenance team to refine **preventive maintenance tasks**, incorporating feedback from operators and technicians.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)